The examination is being carried out on the **following application documents**

Description, Pages

1-43, 45, 46, 48-57, 59, 60, 62-65, 67, 68, 71-83	as published	
44, 47, 58, 61, 66, 69, 70	filed in electronic form on	20-11-2024

Claims, Numbers

1-15	filed in electronic form on	09-05-2024

Drawings, Sheets

1/35-13/35, 16/35, 17/35, 19/35-30/35, 32/35-35/35	as published	
14/35, 15/35, 18/35, 31/35	filed in electronic form on	20-11-2024

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 2018/293809 A1 (JAMES MICHAEL R [US] ET AL) 11 October 2018 (2018-10-11)

D2 EP 3 572 939 A1 (TTTECH AUTO AG [AT]) 27 November 2019 (2019-11-27)

1 Article 56 EPC

Independent claim 1

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 does not involve an inventive step within the meaning of Article 56 EPC. Document D1 is considered to represent the closest prior art and discloses in the wording of claim 1:

A system, comprising:
a vehicle having a network comprising a plurality of end points

(see Par.52-54; Fig.1; Fig.4:100. Please note also the teachings in the last two sentences of Par.11: The diagnostic information requires sensors etc. in the vehicle. These sensors are considered to equate the feature "end points").

The subject-matter of claim 1 therefore differs from this known system by the following distinguishing features:

a controller, comprising:
(i) an automation definition circuit structured to interpret an automation trigger description and an automation action description;
(ii) an automation management circuit structured to provide a trigger detection plan and an automated action plan in response to the automation trigger description and the automation action description; and
(iii) an automation execution circuit structured to determine a trigger event value in response to the trigger detection plan, and to provide an automation command in response to the trigger event value and the automated action plan; wherein
(iv) an end point of the plurality of end points is responsive to the automation command to implement an automated vehicle response.

and is thus new.

The problem to be solved by the present invention may therefore be regarded as to cost effectively update functions of the vehicle, cf. Par.7, last sentence.

The solution proposed in claim 1 of the present application cannot be considered to involve an inventive step (Articles 52(1) and 56 EPC) for the following reasons:

From document D2 a method for updating software functions in a vehicle is known, see eg. Par.1.

In D2, a controller controls automotive applications and updates (see eg. Par.18, Par. 33) according to an "automation trigger description" in the form of scheduled tasks ("=trigger detection plan") which are "triggered" to be executed, see Fig.2.

These tasks in D2 can be considered as "operations as part of a workflow" and thus correspond to the example in the description of the present application in eg. Par.166 for the feature "automation action description". The timing diagram in Fig.2 is considered to equate the feature "trigger detection plan".

It follows that the distinguishing features (i) and (ii) are disclosed in D2.

The distinguishing feature (iii) is also disclosed in D2, please consider the execution of the scenario which is shown in Fig.2: An "automation execution circuit" (which is considered to be implicitly disclosed in Par.35) determines a "trigger event value" (here: a data indicating which task is when applied to which core) in response to the "trigger detection plan". The execution of a corresponding task at the scheduled time according to Fig.2 is considered to equate the feature "provide an automation command".

The distinguishing feature (iv) is also disclosed in D2: The "automated vehicle response" is the storage of the diagnostic information. Such interpretation is in line with one of the examples of the feature "automated vehicle response activity" which is provided by the applicants in Par.152 in the description.

When solving the technical problem the skilled person would have no practical difficulties in applying these teachings from D2 to the system which is disclosed in D1, thus arriving at the subject-matter of claim 1. The skilled person would thus arrive at the subject-matter of claim 1 without exercising any inventive skill (A.56 EPC).

As a consequence, the subject-matter of claim 1 is not inventive (Article 56 EPC).

2 Dependent claims

The dependent claims do not appear to contain any additional features which, in combination with the features of any claim to which they refer, are novel (A.54 EPC) or involve an inventive step (A.56 EPC) because these claims concern commonplace features which are either derivable from the documents cited in the Search Report or obvious to a skilled person. For example:

The subject-matter of claims 2,3 defines a "triger evaluation command" and its processing. Such commands process data from sensors to check if certain conditions are met, see eg. Par.167. However, such subject-matter appears to be well-known and represent obvious design options to ensure correct operation and thus cannot contribute to inventive step (not inventive).

The subject-matter of claims 4,5 is disclosed in D2, Fig.2-3 (not inventive).

The subject-matter of claims 6-15 provides definitions for the features "automated vehicle response", "automation action description", "automated action plan" which definitions are considered to represent design options the skilled person would apply under the dictate of circumstances and in view of D2, Par.38-40 (not inventive).

3 Miscellaneous

It is not at present apparent which part of the application could serve as a basis for a new, allowable claim. Should the applicant nevertheless regard some particular matter as patentable, an independent claim should be filed taking account of Rule 43(1) EPC. The applicant should also indicate in the letter of reply the difference of the subject-matter of the new claim vis-à-vis the state of the art and the significance thereof.

Should the applicants decide to file new claims, the following should be observed:

In order to assure the economy and efficiency of the procedure, it is requested that the following additional points be dealt with **immediately**:

The applicant **should adapt the description to the claims on file** and delete or amend any statements or expressions which throw doubt on the scope of protection (GL F-IV, 4.3). Failure to do so could lead to the refusal of the application under Article 97(2) EPC in combination with Article 84 EPC.

The applicant should amend the description as needed to fulfil the following requirements: The technical field of the invention (Rule 42(1)(a), GL F-II.4.2) must correspond to the amended set of claims.

The summary of the invention (Rule 42(1)(c) and GL F-II.4.5) must correspond to the amended set of claims. If appropriate, statements such as “The invention is set out in the appended set of claims“ or "The invention is as defined in claim X" can be applied instead of repeating the claims verbatim.

All embodiments which are not covered by the claims any more must be deleted, unless these embodiments can reasonably be considered to be useful for highlighting specific aspects of the amended subject-matter (Guidelines F-IV, 4.3).

All technical features of the independent claims shall not be presented as optional in the description. Terms such as “for example“, “may“, “can”, “exemplary“, “optionally“, “preferably“ and the like should be removed when preceding a feature of an independent claim.

To meet the requirements of Rule 42(1)(b) EPC, the documents D1 and D2 should be identified in the description and the relevant background art disclosed therein should be briefly discussed.

The paragraph 251 of the present description, referring to the "Spirit of the invention", should be brought into accordance with the Guideline F-IV, 4.4.

The paragraph 4 "Incorporated by reference", should be cancelled according to Guidelines F-III, 8.

Furthermore:

To meet the requirement of Rule 43(1) EPC any new independent claim should be properly cast in the two part form, with those features which in combination are part of the prior art (see document D1) being placed in the preamble.

Reference signs in parentheses should be inserted in the claims to increase their intelligibility, Rule 43(7) EPC. This applies to both the preamble and characterising portion (see the Guidelines, F-IV, 4.18).

The applicant is requested to file amendments by way of replacement pages, avoiding unnecessary recasting of the description. Where replacement pages are filed and it is not obvious how the text has been amended, the applicant should explain from which points of the original application documents the amendments are derived (Guidelines H-III, 2.2). It should be clear from the reply on which original claims each new claim is based, whether and how the claim has been amended and where the basis for each amendment (which can result from adding, amending or removing features) can be found.

To deal with the procedure in the most constructive and efficient way, the applicants are requested to provide a **full complete set** of application documents, i.e. both amended and not amended parts (description, claims and drawings).